

Online sales Business Analysis

Objective:

This analysis constructs a complete visual model of the business through an interactive dashboard, enabling stakeholders to explore and understand the business as an integrated whole.

From this general view, specific aspects were carefully selected for deeper exploration and to address key questions.

Finally, meaningful insights were extracted from this focused exploration to support clear, data-driven business decisions.

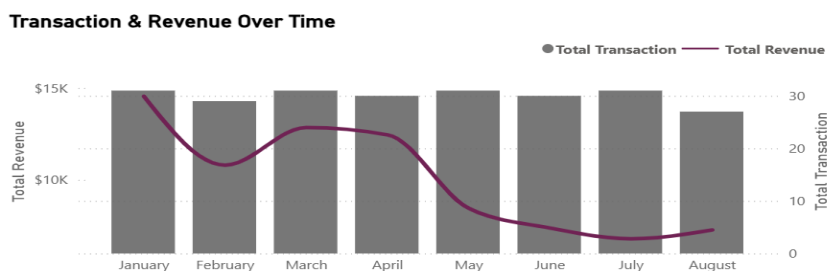
Data overview:

The dataset used in this analysis is a clean, structured sales dataset sourced from Kaggle. It consists of a single table capturing online sales transactions over a period of eight months. Each record represents an individual transaction, including details on the product, price, total revenue and geographic region.

Key Observations and Insights:

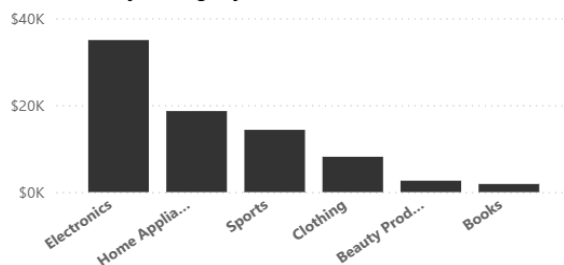
Based on the dashboard, The data show that:

1. Total transactions remain stable over the 8 months, but total revenue is steadily declining. This means we are not making the most out of each transaction, which indicates the existence of a value gap.

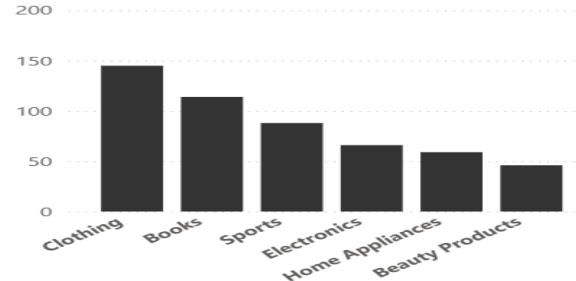


- Products with high revenue have the smallest sales volume, while low-revenue products take up most of our sales volume. This means we are focusing our effort on low-value items, which results in an efficiency imbalance.

Revenue By Category



Units Sold By Product Category



- High-volume items like Books and Clothing (\$13 & \$57) sell 259 units but only make \$10k. In contrast, Electronics (\$399) makes \$34.9k from only 66 units. This means there is an inappropriate distribution of prices, and we are missing a major opportunity to capture more value from our high-traffic items.

Category Performance Details

Product Category	Median of Unit Price	Units Sold	Total Revenue
Electronics	\$399	66	\$34,982.41
Home Appliances	\$200	59	\$18,646.16
Sports	\$148	88	\$14,326.52
Clothing	\$57	145	\$8,128.93
Beauty Products	\$38	46	\$2,621.9
Books	\$13	114	\$1,861.93
Total	\$90	518	\$80,567.85

The strategy: Optimize Product Price

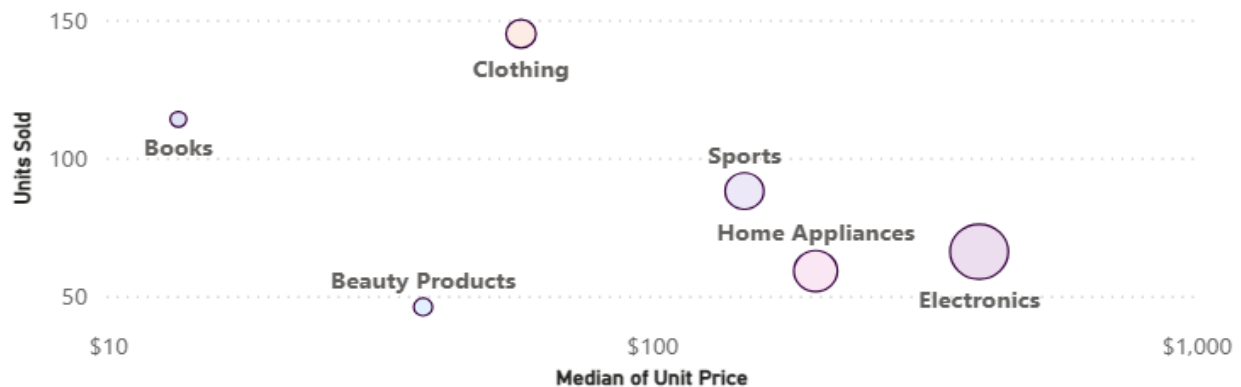
To address the decline in revenue, we must close the Value Gap while simultaneously increasing the number of transactions. This gap is created by the interaction between the Number of Units Sold and the Unit Value.

Our strategy focuses on optimizing Product Pricing as the Primary lever to influence both demand and marketing effectiveness. By aligning our prices with the specific characteristics of each category, we create more value for the business:

- Books: Habitual and non-seasonal. Since demand is steady, they can support a price increase to capture more value from high-volume traffic.

- Clothing & Sports: seasonal and needs-based. Prices should be adjusted to match seasonal demand patterns to maximize turnover.
- Electronics & Home Appliances: High-value, infrequent purchases. These require targeted marketing in wider regions to increase the number of high-value interactions.
- Beauty Products: Consumable and recurring. Pricing should encourage repeat purchases to maintain steady revenue.

Category Performance



Specific Recommendations:

Based on the Category Performance chart and product characteristics, here is the direct plan for each category:

- Books: Because demand is steady, we suggest a 10% price increase to capture more value from every transaction.
- Clothing: To maximize our frequent sales, we recommend a 5% price increase.
- Sports: These are seasonal items with moderate volume and pricing. We suggest a 5% price increase specifically during peak seasons to take advantage of higher demand.
- Beauty Products: These are consumable items where we need to increase the number of units sold per customer. We recommend maintaining the current price while using bundle deals to move these products higher on the “Units Sold” axis.
- Electronics & Home Appliances: These are high-value, “one-time” purchases that already have high prices. We recommend a 0% price change (no increase) to avoid hurting demand; instead, focus on marketing to reach wider regions.

I suggest the business owner apply these exact percentages (+10% for books and +5% for clothing) using the What-if analysis slider. This will show the immediate rise in total revenue while assuming our existing transaction patterns remain the same.